

7 The truth table for a logic circuit is shown.

INPUT				OUTPUT
A	B	C	D	T
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

(a) Write the Boolean logic expression that corresponds to the given truth table as the sum-of-products.

T =
..... [3]

(b) Complete the Karnaugh map (K-map) for the given truth table.

AB					
CD		00	01	11	10
	00				
	01				
	11				
	10				

[2]

(c) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

(d) (i) Write the Boolean logic expression from your answer to part (c) as the simplified sum-of-products.

T =
..... [2]

(ii) Use Boolean algebra to write your answer to part (d)(i) in its simplest form.

T = [1]